

Transport Layer in IoT Communications

Internet of Things



PROF.: CWILLIAMS@UTEC.EDU.PE

SRC: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Zach Shelby and Carsten Bormann. 2010: The Wireless Embedded Internet. Wiley Publishing.

Executive Summary

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- **Motivation:** Network implementation is half the work of an IoT implementation.
- **Problem:** Recalling basic concepts of networking
- **Topics:**
 - Basic concepts review
 - Transport Layer
 - Transport Layer applied to IoT sensor network

Outline

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Introduction

UDP

TCP

IoT Application

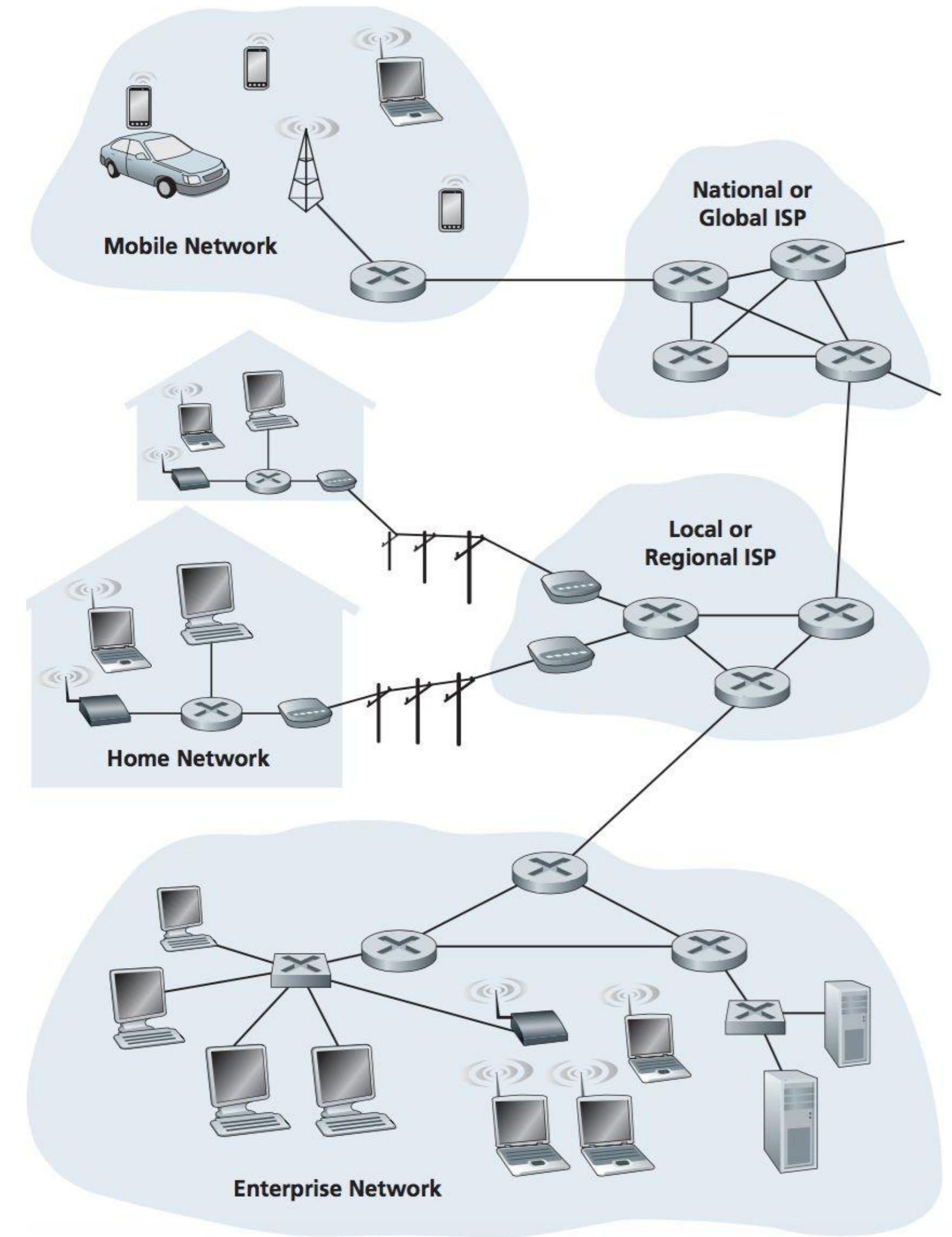
Conclusion

What is the Internet?

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A components view:

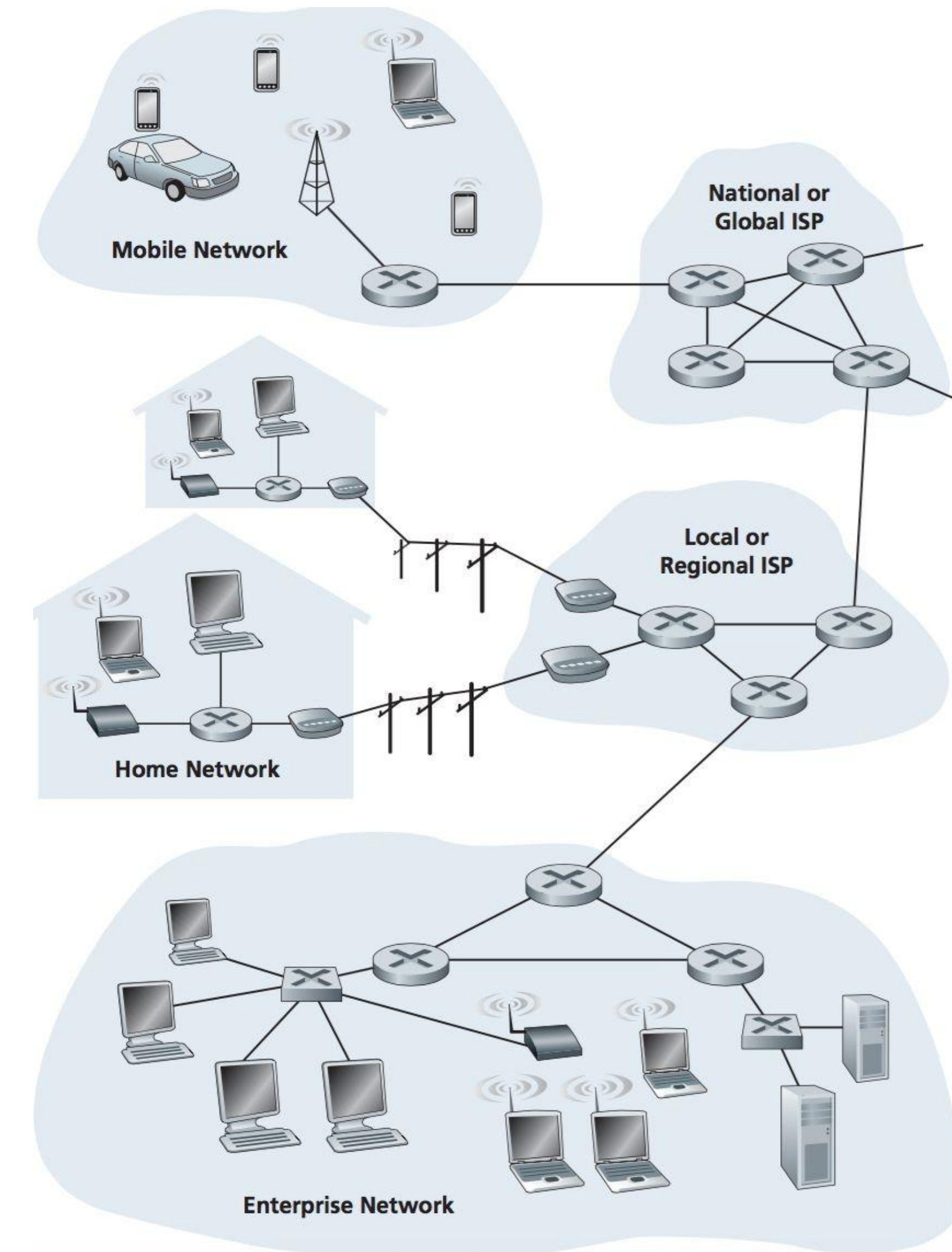
- Millions of connected computing devices:
 - Hosts (end systems) running network apps
- Communication links:
 - Fiber, copper, radio, satellite
 - Transmission rate: bandwidth
- Packet switches
 - Forward packets (chunks of data)
 - Routers and switches



What is the Internet?

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- The Internet is all about connecting end systems to each other
- End systems access the Internet through Internet Service Providers (ISPs):
 - Residential ISPs, corporate ISPs, university ISPs
- Each ISP is a network => Internet: "network of networks"
- Protocols control sending and receiving of messages
- Internet standards:
 - IETF (Internet Engineering Task Force)
 - RFC: Request for Comments (ex. [TCP RFC](#))



What is the Internet?

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A service view:

- Infrastructure that provides services to applications: Web, VoIP, email, games, social networks, e-commerce, ...
- Provides programming interfaces to applications
 - Hooks that allow a program running on one end system asks to deliver data to a specific destination program running on another end system.
 - Provides service options, analogous to the postal service

Modelo OSI

7. Aplicación

6. Presentación

5. Sesión

4. Transporte

3. Red

2. Enlace de datos

1. Física

Modelo TCP/IP

Aplicación

Transporte

Internet

Acceso a la red

What is a protocol?

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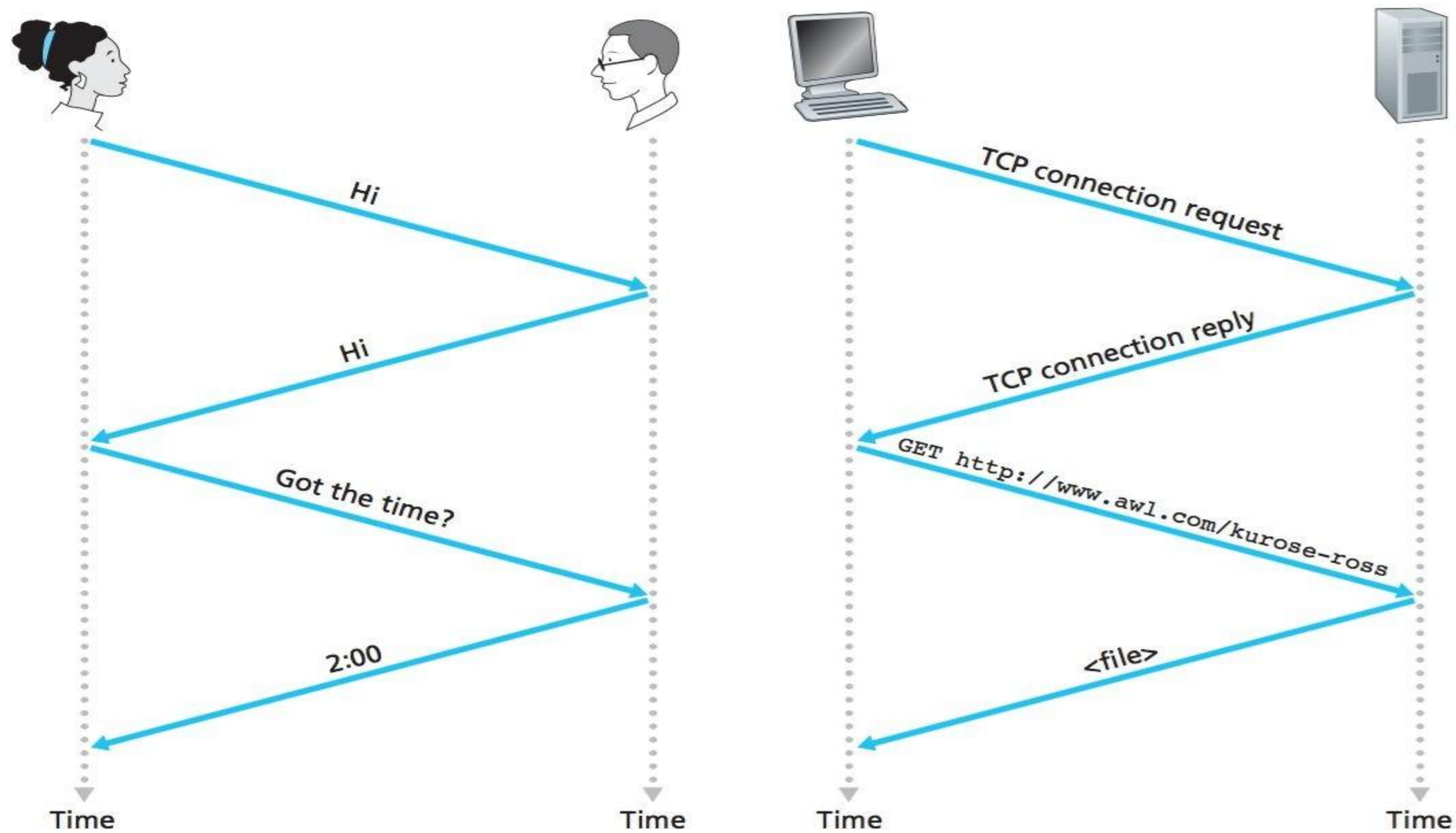


Figure 1.2 ♦ A human protocol and a computer network protocol

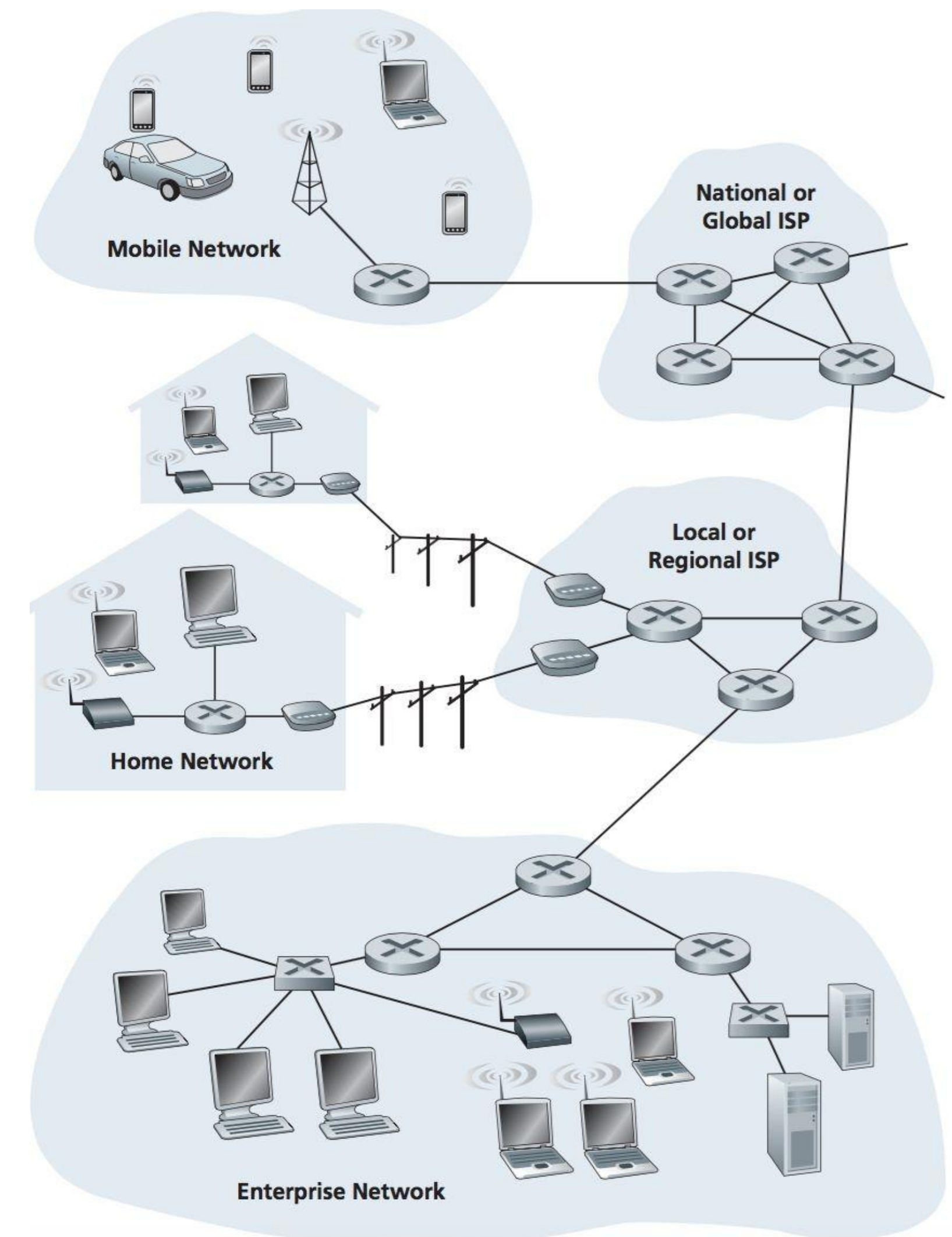
What is a Protocol?

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*A **protocol** defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event*

A closer look at network structure

- Network edge
 - Hosts = end systems: clients and servers
 - Servers often in data centers
- Access networks
 - Physical media: wired or wireless communication links
 - A local area network (LAN) is usually used to connect an end system to the edge router.
- Network core
 - Interconnected routers
 - Network of networks



Data transmission

1. Host takes application message, breaks into smaller chunks (**packets**), and transmits them into access network
2. Through the access network packets reach the core network
3. Core network forwards packets from one router to the next, across links on path from source to destination

Packet switching: store and forward.

Encapsulation

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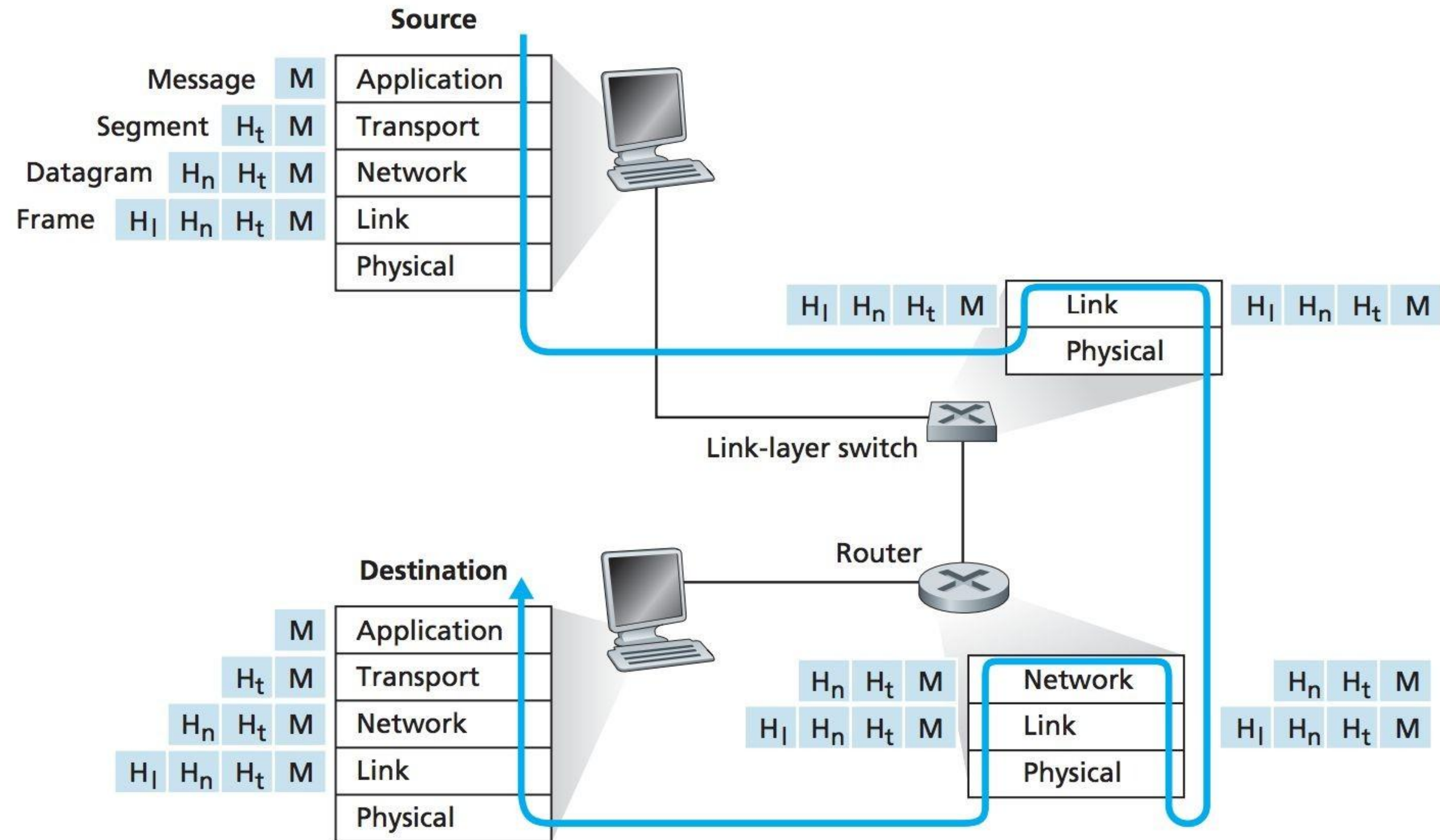
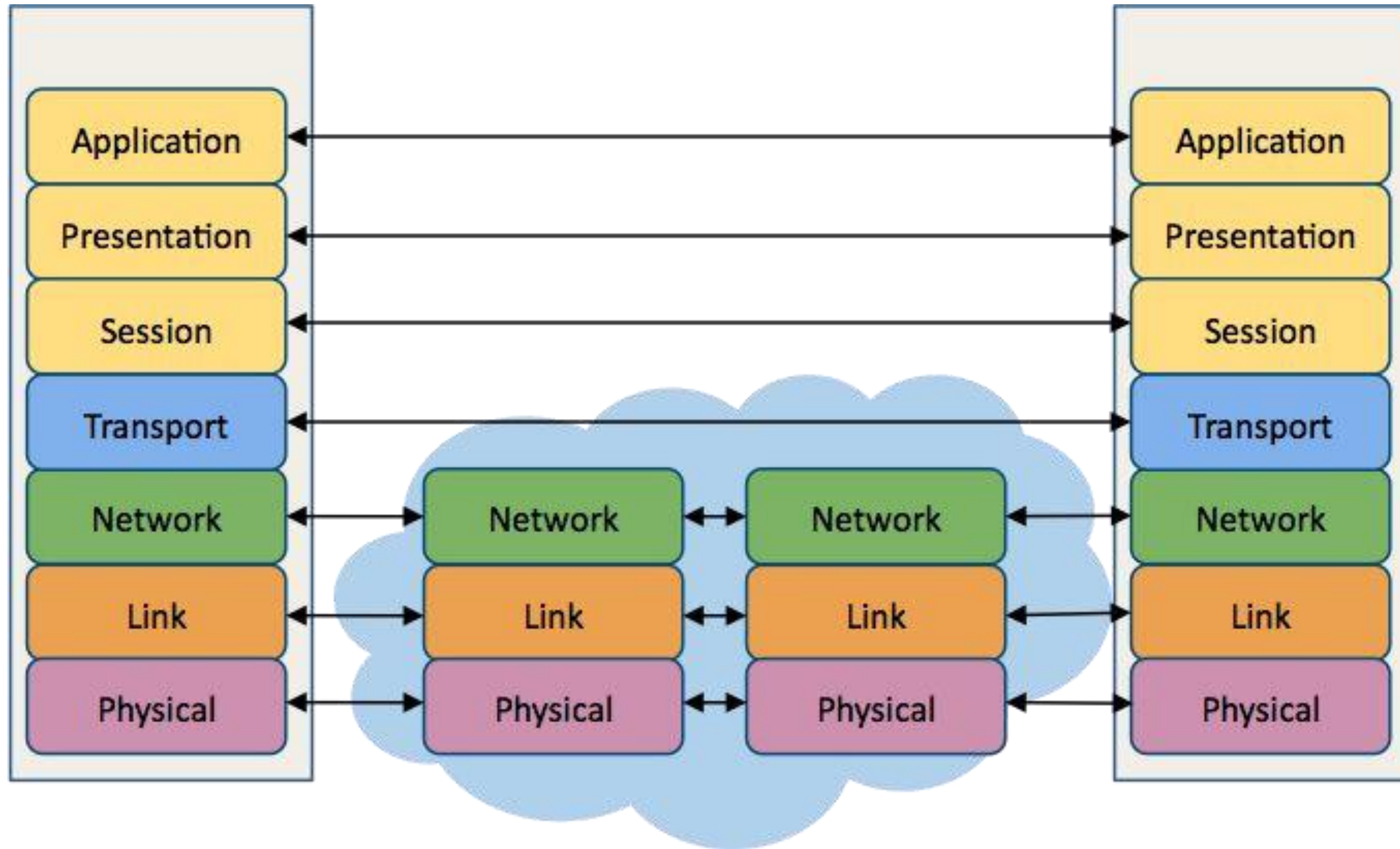


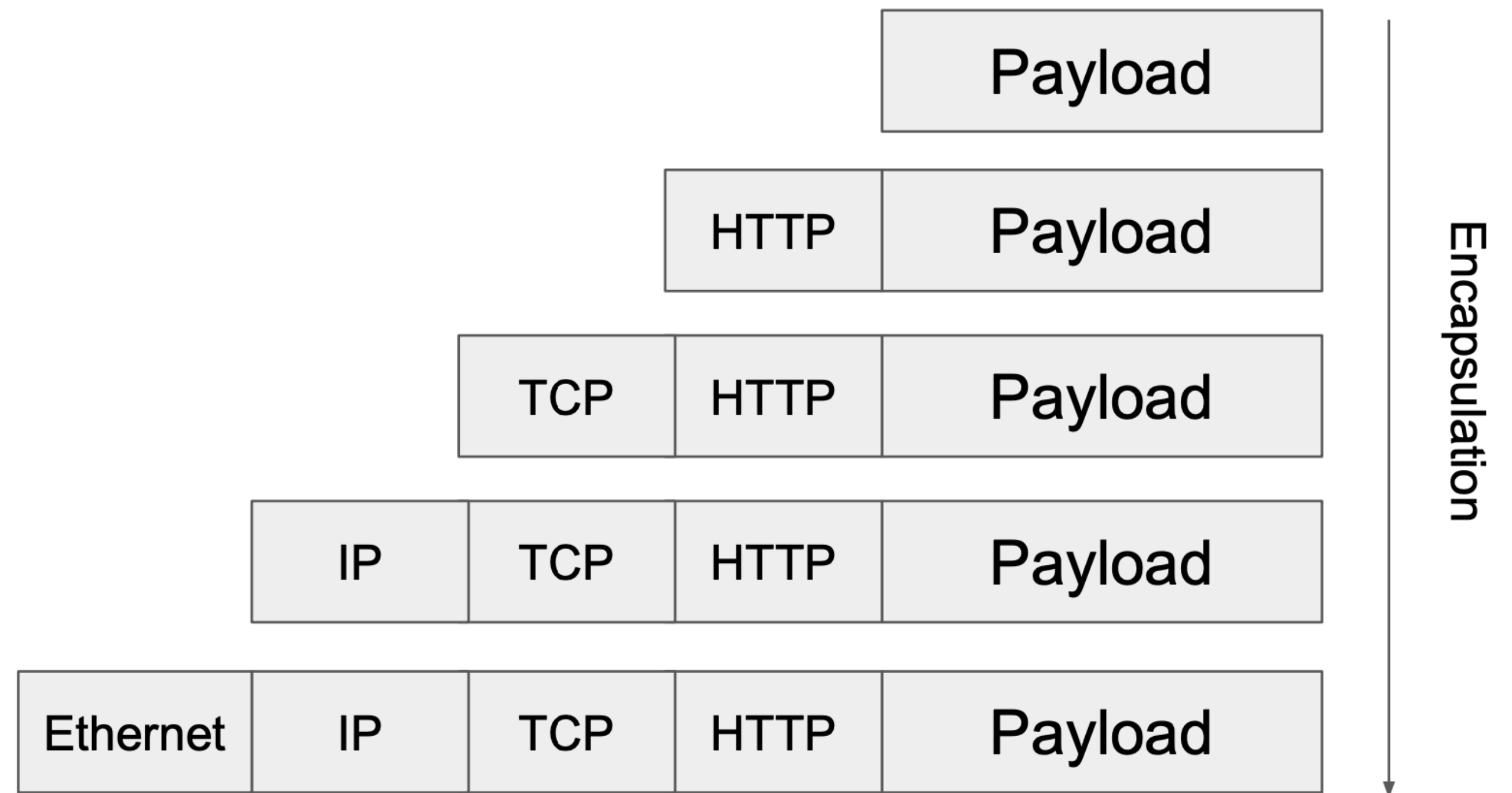
Figure 1.24 ♦ Hosts, routers, and link-layer switches; each contains a different set of layers, reflecting their differences in functionality

Transport protocols



Recall: Encapsulation

- Example:
 - An HTTP message
 - Match The Layers to the Example. Discuss



Transport protocols

- End-to-end communication
- Communication between software application processes on different hosts.
- Multiplex/demultiplex
- Abstracts network details for the applications
- Abstracts different applications for the network
- Main protocols: UDP and TCP

Transport protocols

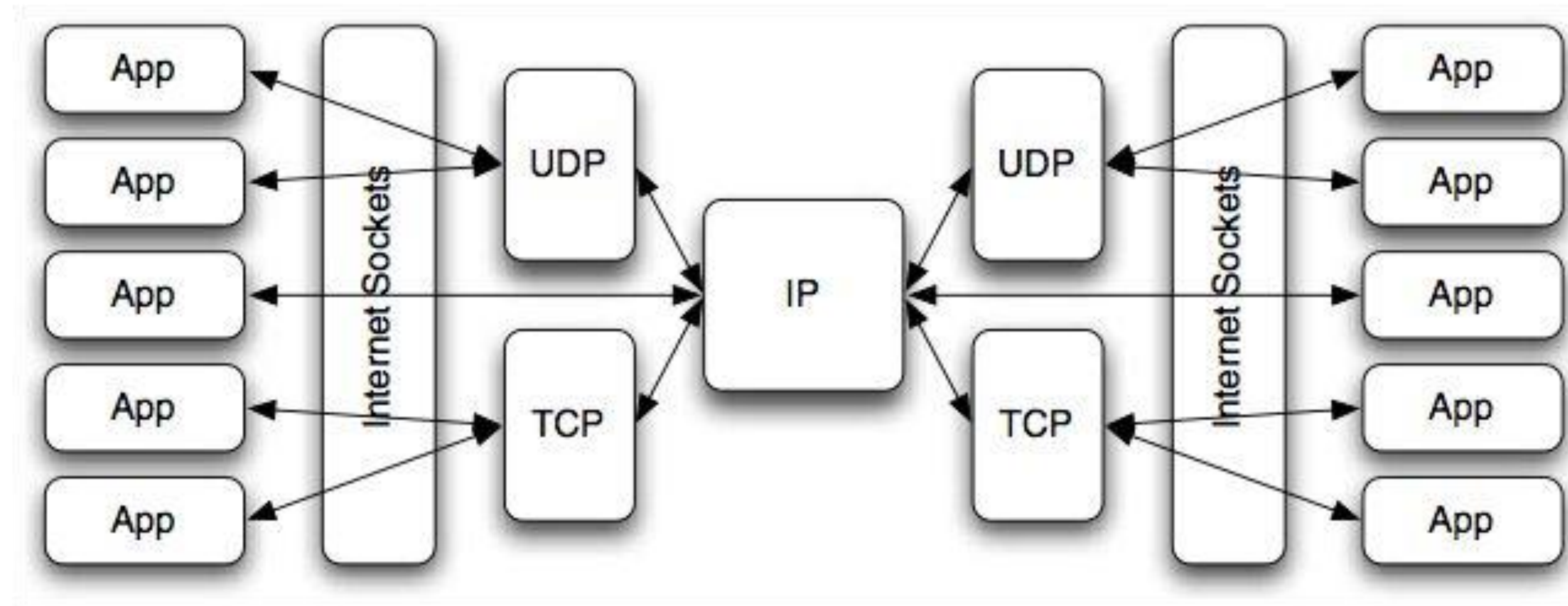


Figure 5.1 Applications process communication occurs through Internet sockets.

Source: Zach Shelby and Carsten Bormann. 2010: The Wireless Embedded Internet. Wiley Publishing.



Packets Creation



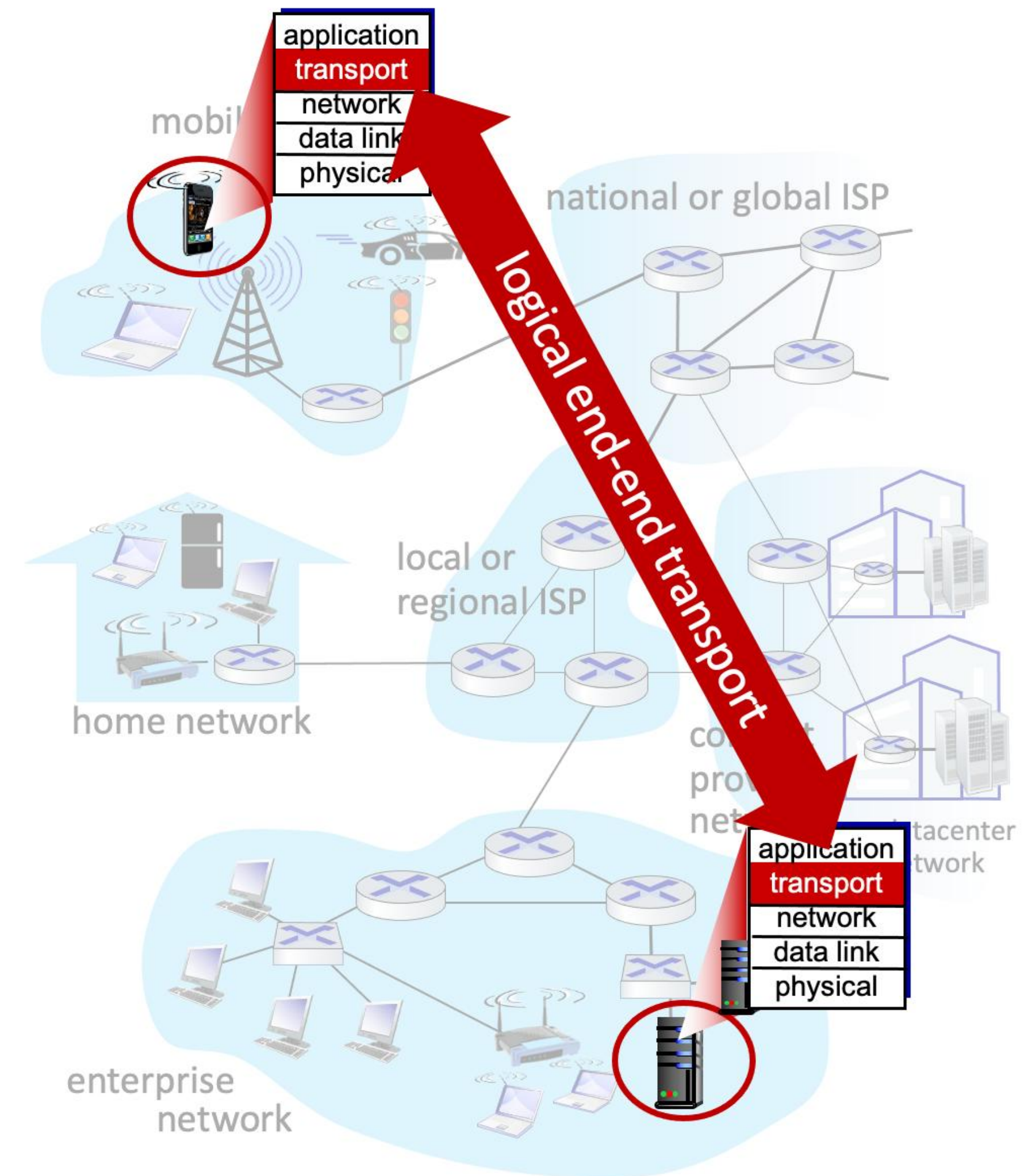
Transport



Packets Assembly

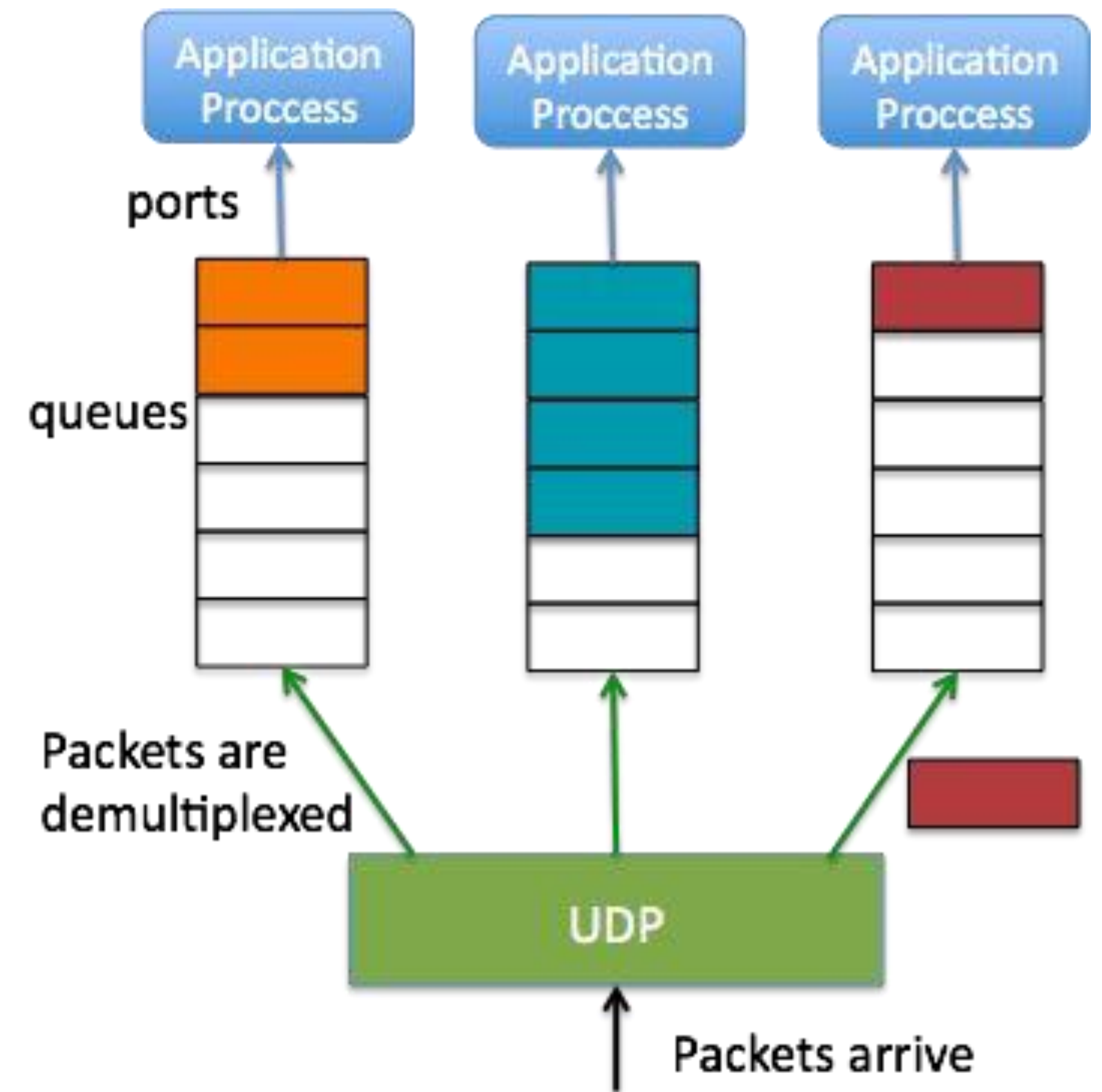
Transport services and protocols

- provide *logical communication* between application processes running on different hosts
- transport protocols actions in end systems:
 - sender: breaks application messages into *segments*, passes to network layer
 - receiver: reassembles segments into messages, passes to application layer
- two transport protocols available to Internet applications
 - TCP, UDP



UDP

- User Datagram Protocol
- Simplest protocol in the TCP/IP suite
- RFC 768
- Per-process addressing by the use of ports (16 bits).



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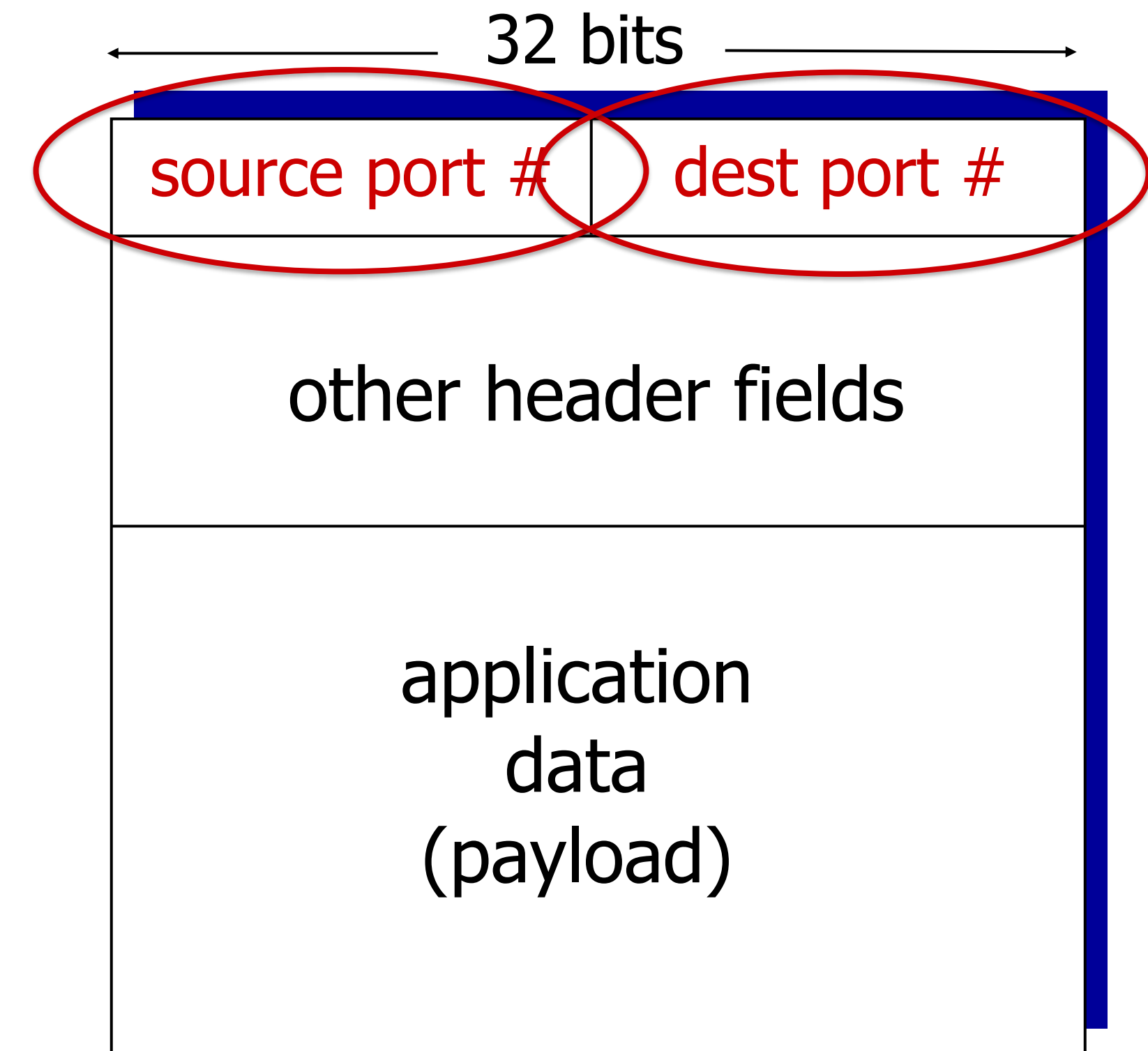
Conclusion

UDP

- Best-effort service:
 - IP network does its best to deliver the datagram;
 - no guarantees that the datagrams are delivered at the destination;
 - no guarantees that the datagrams are delivered in the same order as they were sent.

UDP header

- host receives IP datagrams
 - each datagram has source IP address, destination IP address
 - each datagram carries one transport-layer segment
 - each segment has source, destination port number
- host uses *IP addresses & port numbers* to direct segment to appropriate socket



TCP/UDP segment format

TCP

- Transport Control Protocol
- Reliable byte stream on top of the best-effort packet service provided by the IP layer.
- TCP hides the packet-oriented IP network beneath a virtual circuit abstraction
 - connection
- A connection is identified by the IP addresses and TCP port numbers of the end points.

TCP

- Reliable delivery:
 - Acknowledgments and retransmissions
 - Sequence numbers
 - Sliding window

TCP ACK

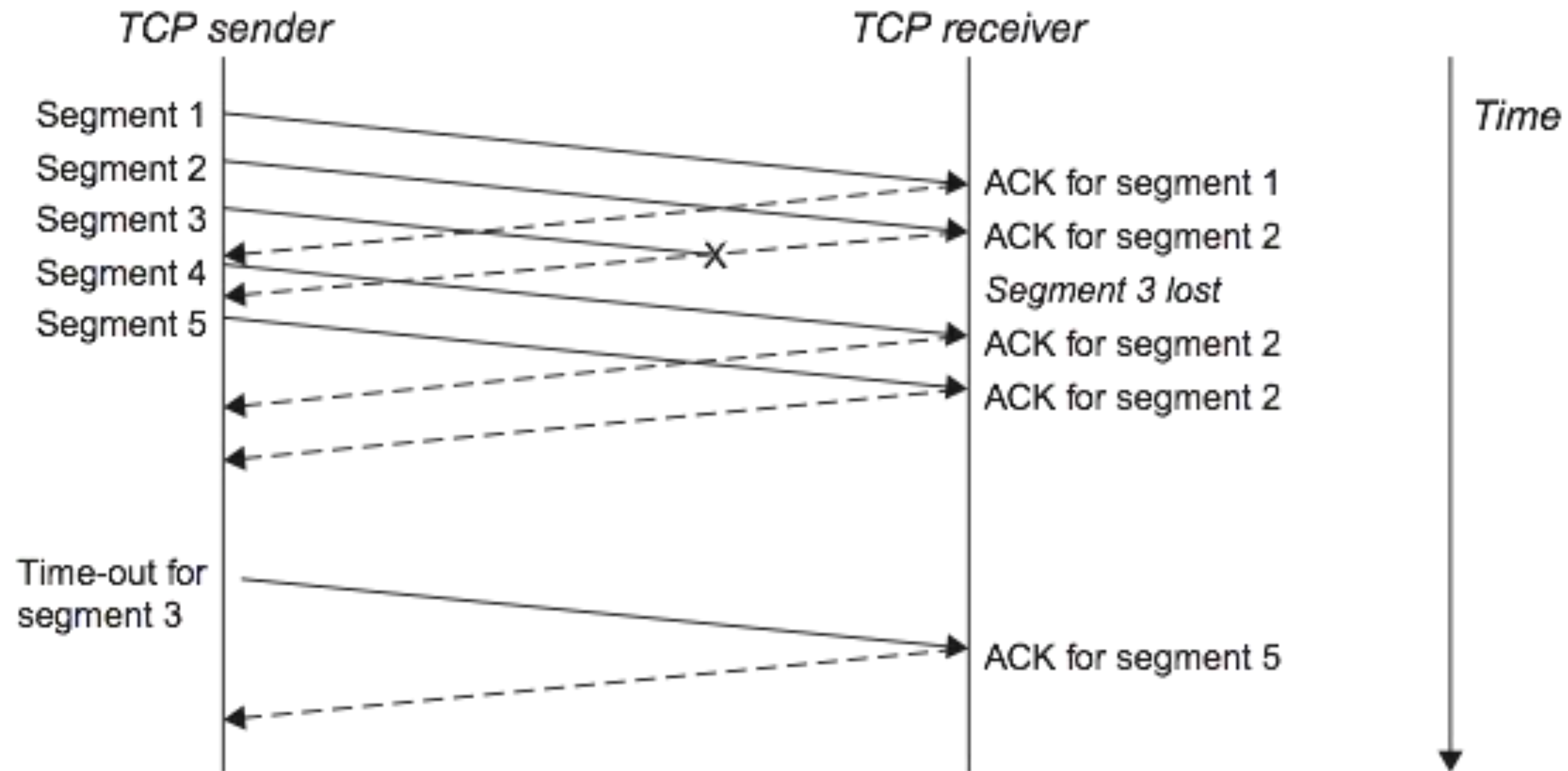
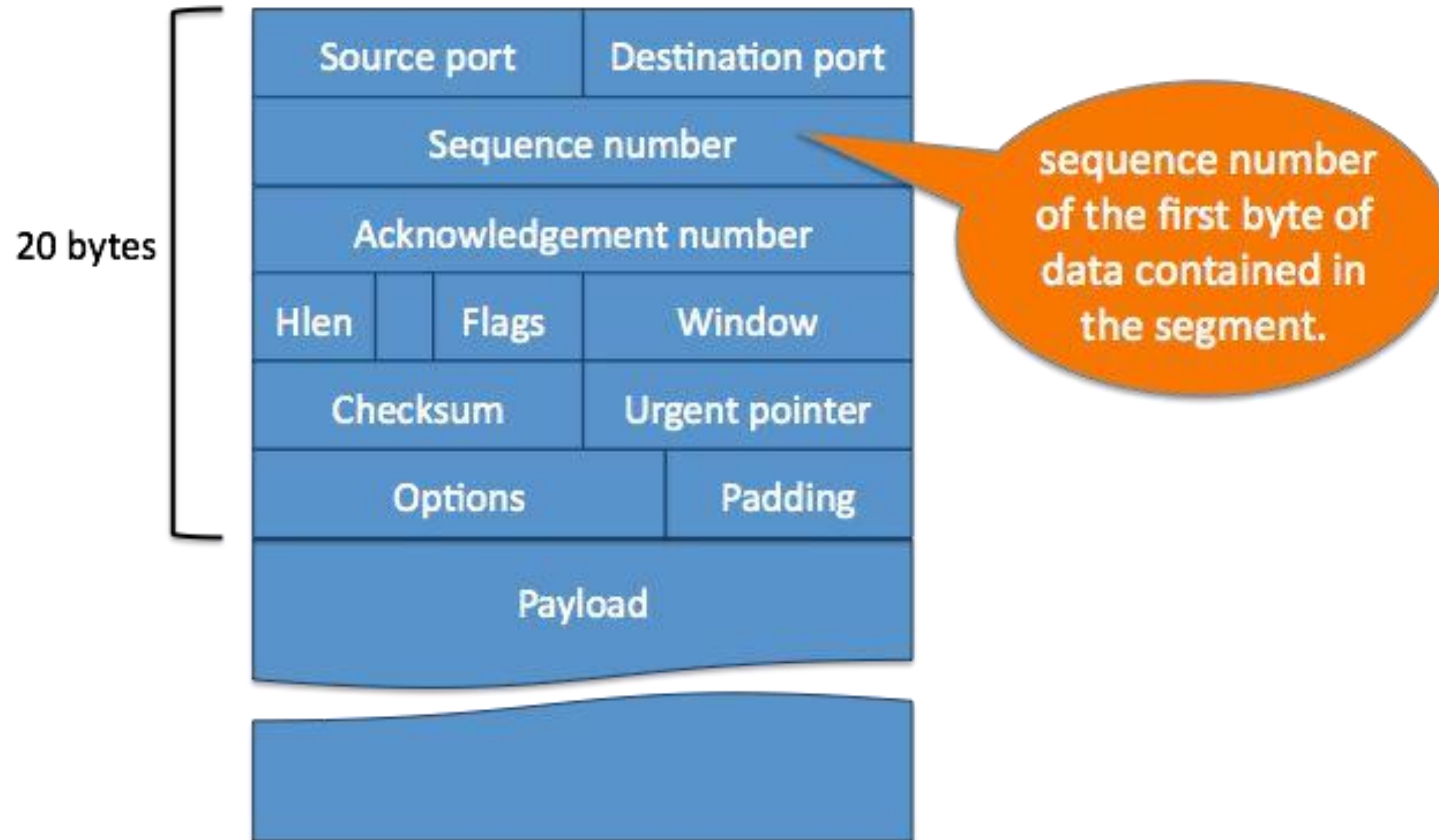


FIGURE 6.4

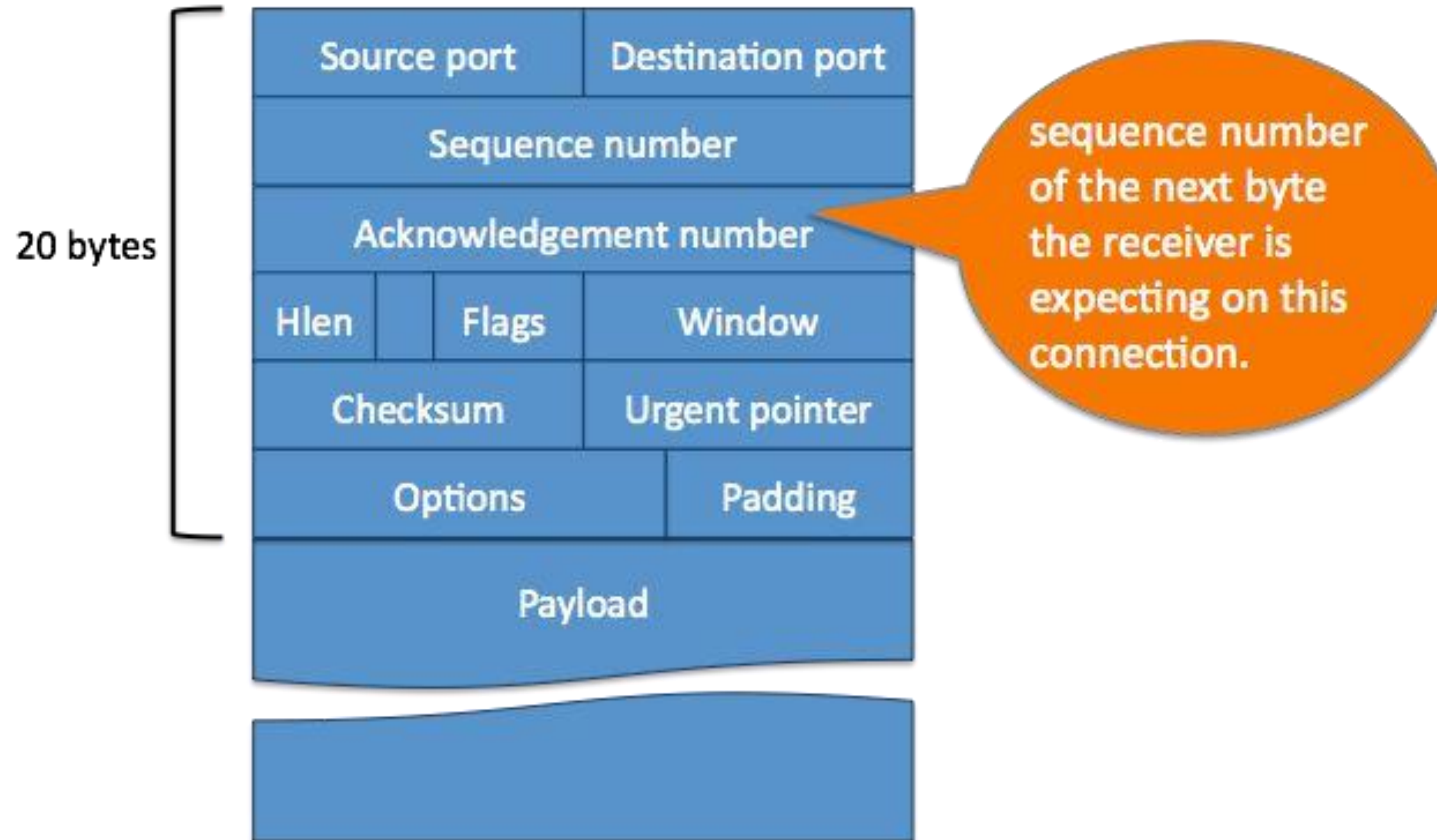
Loss of a TCP segment and the corresponding time-out.

Source: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

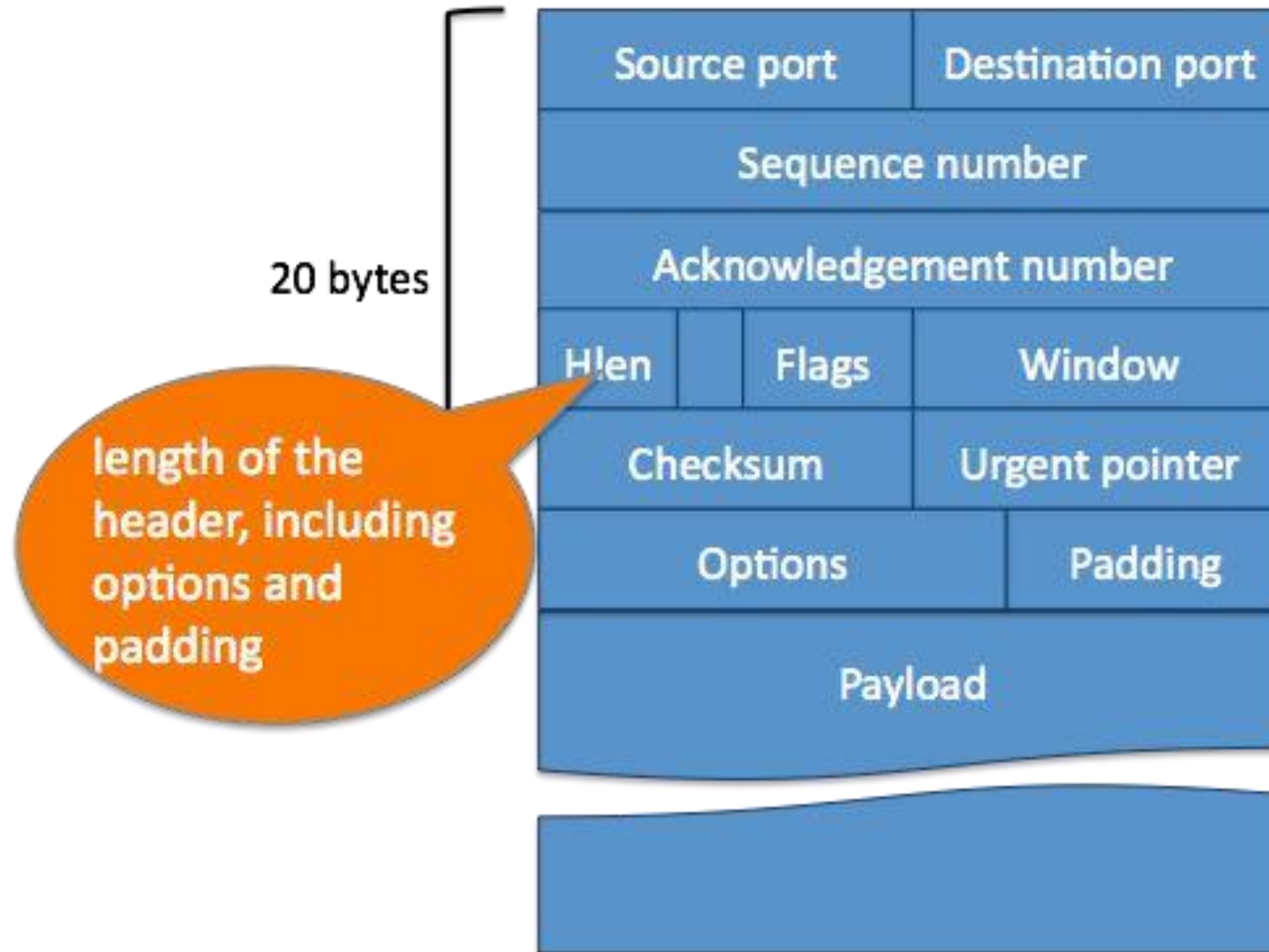
TCP Header



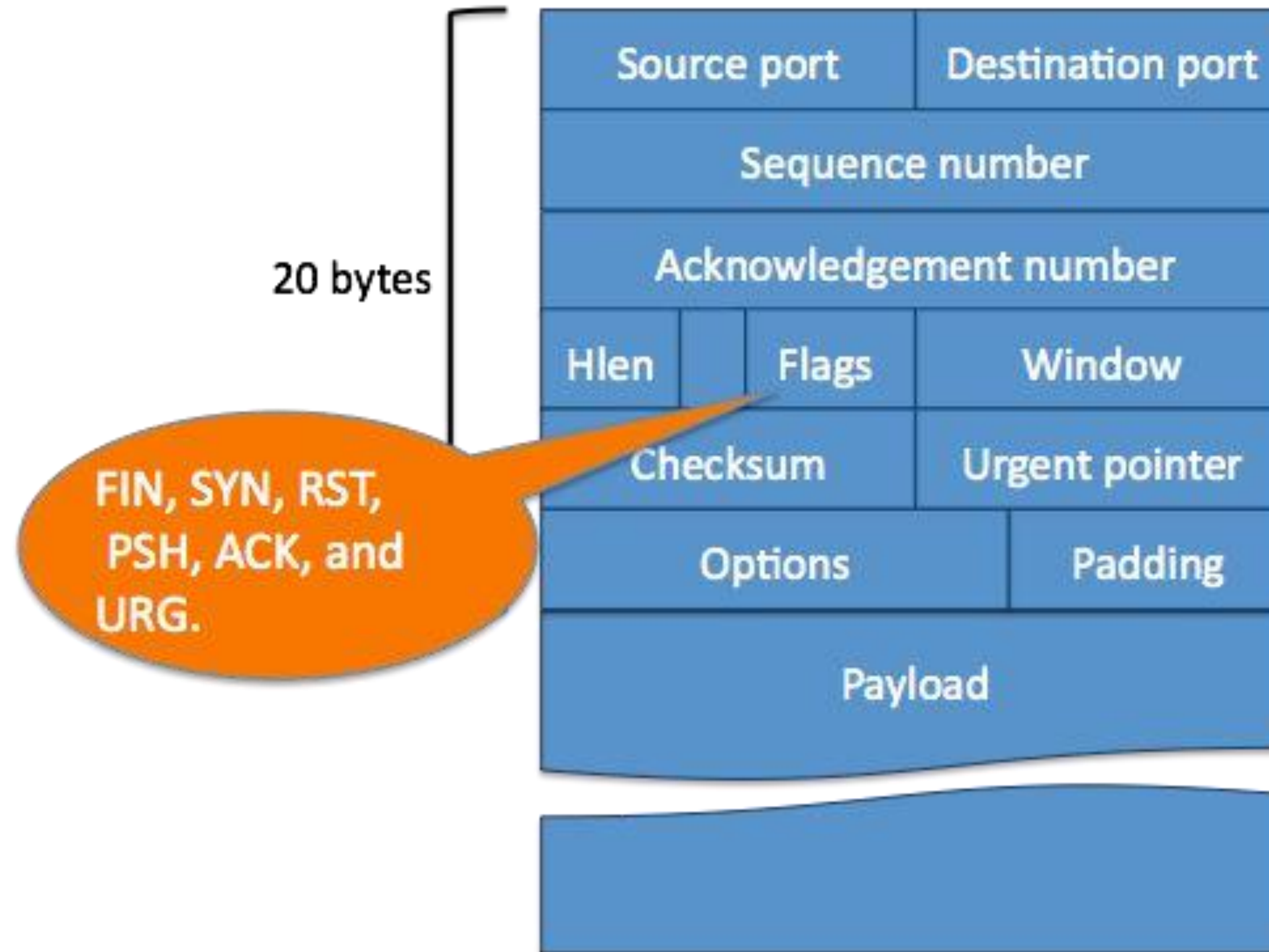
TCP header



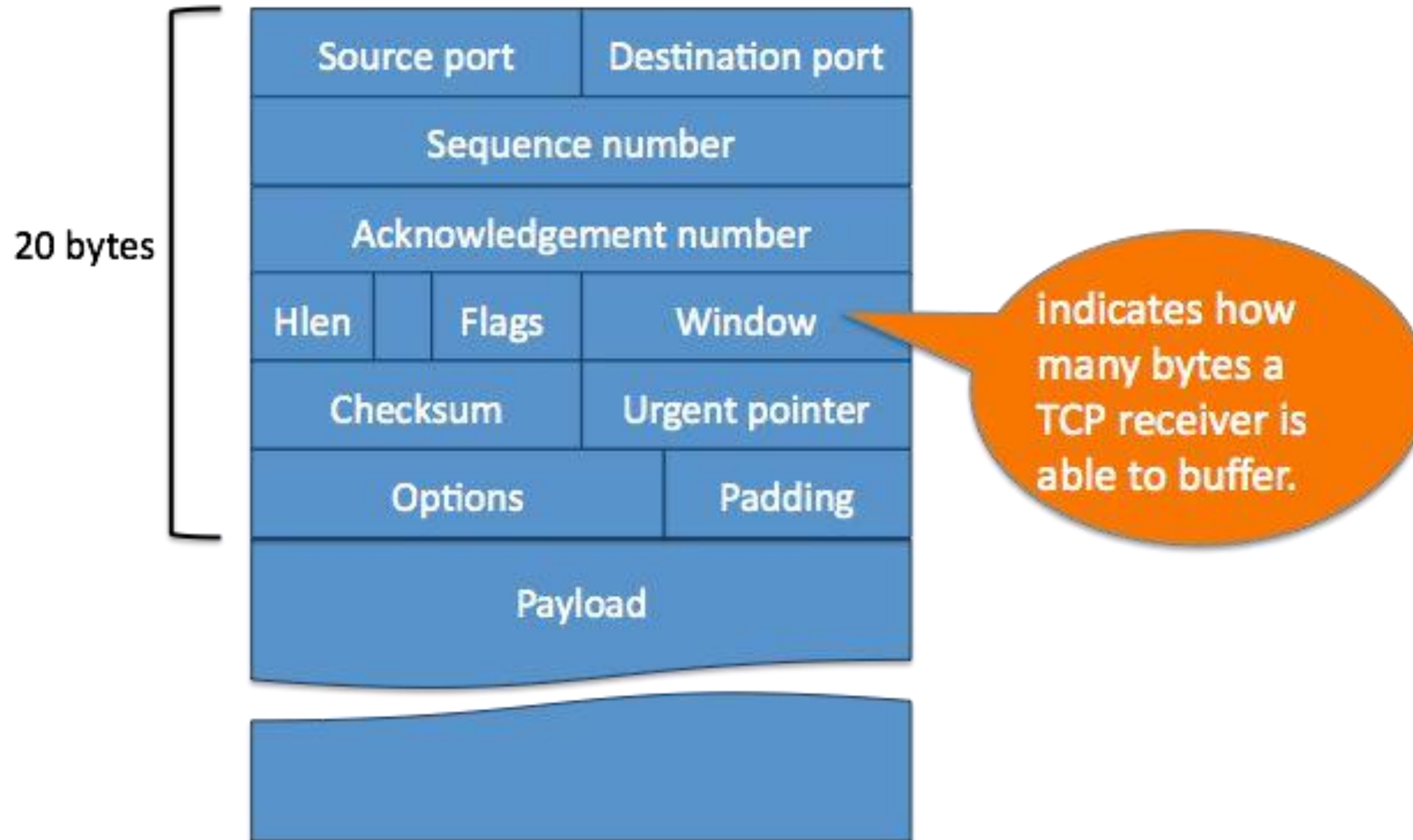
TCP Header



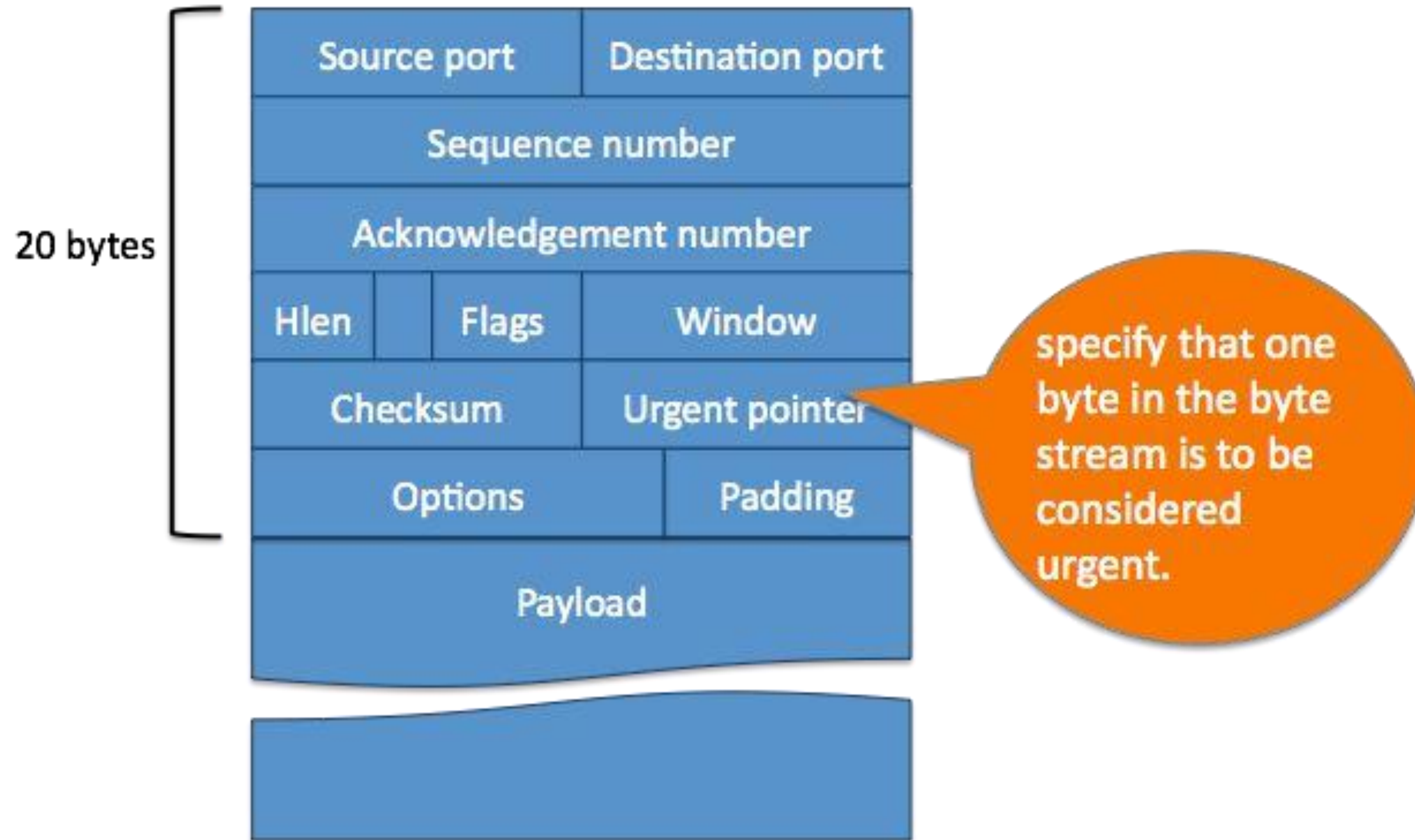
TCP Header



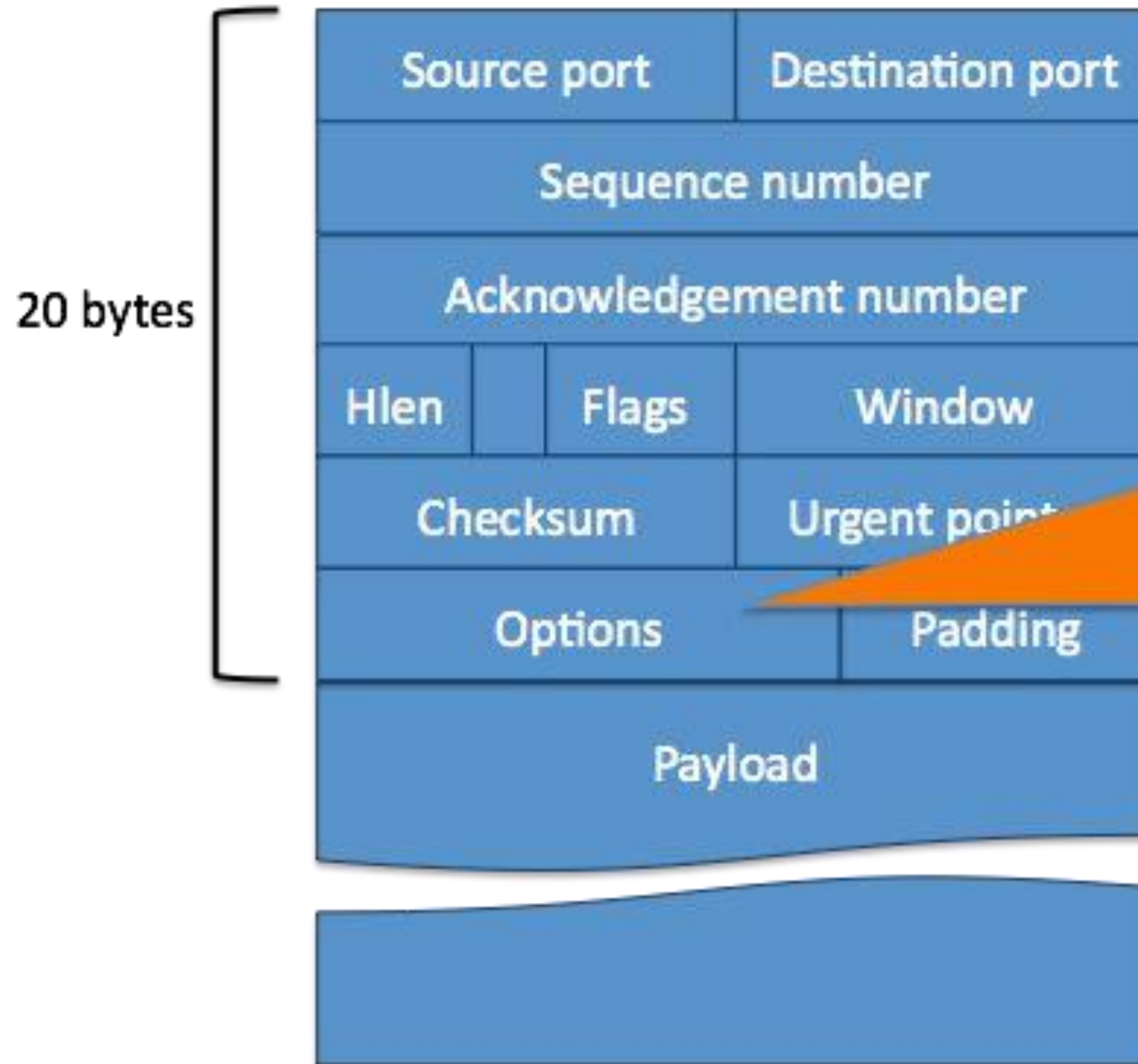
TCP Header



TCP Header



TCP Header



MSS option specifies the largest TCP segment size that a TCP end point is able to accept.

TCP three-way handshake

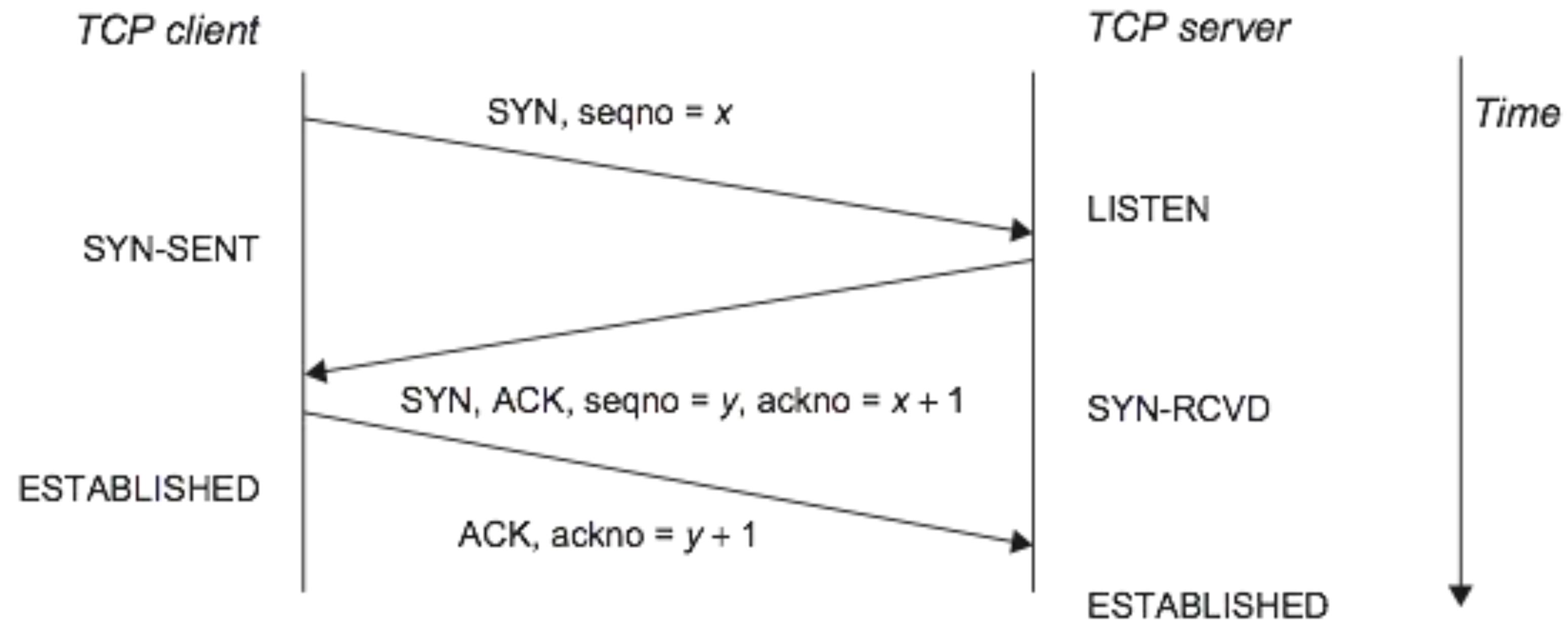


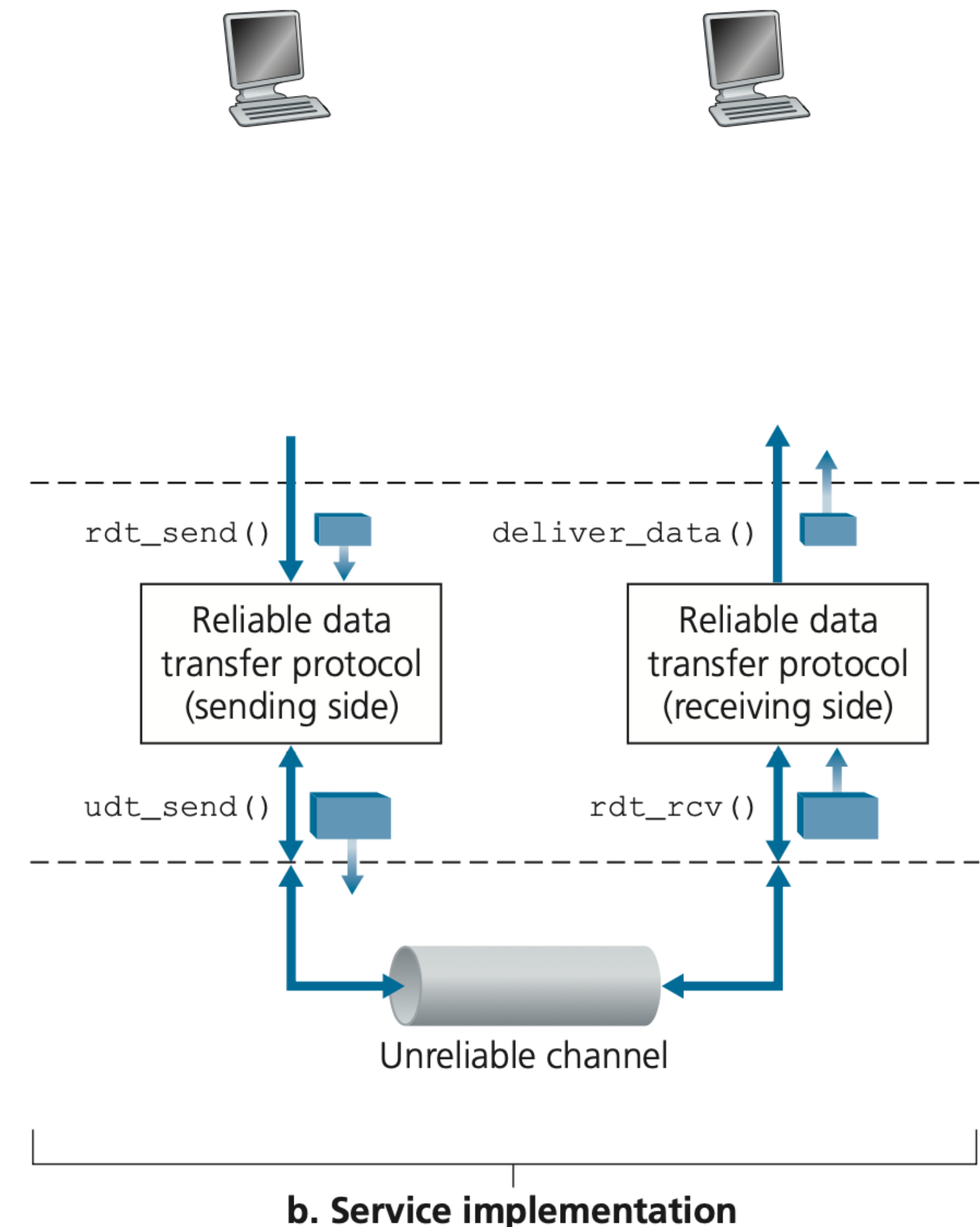
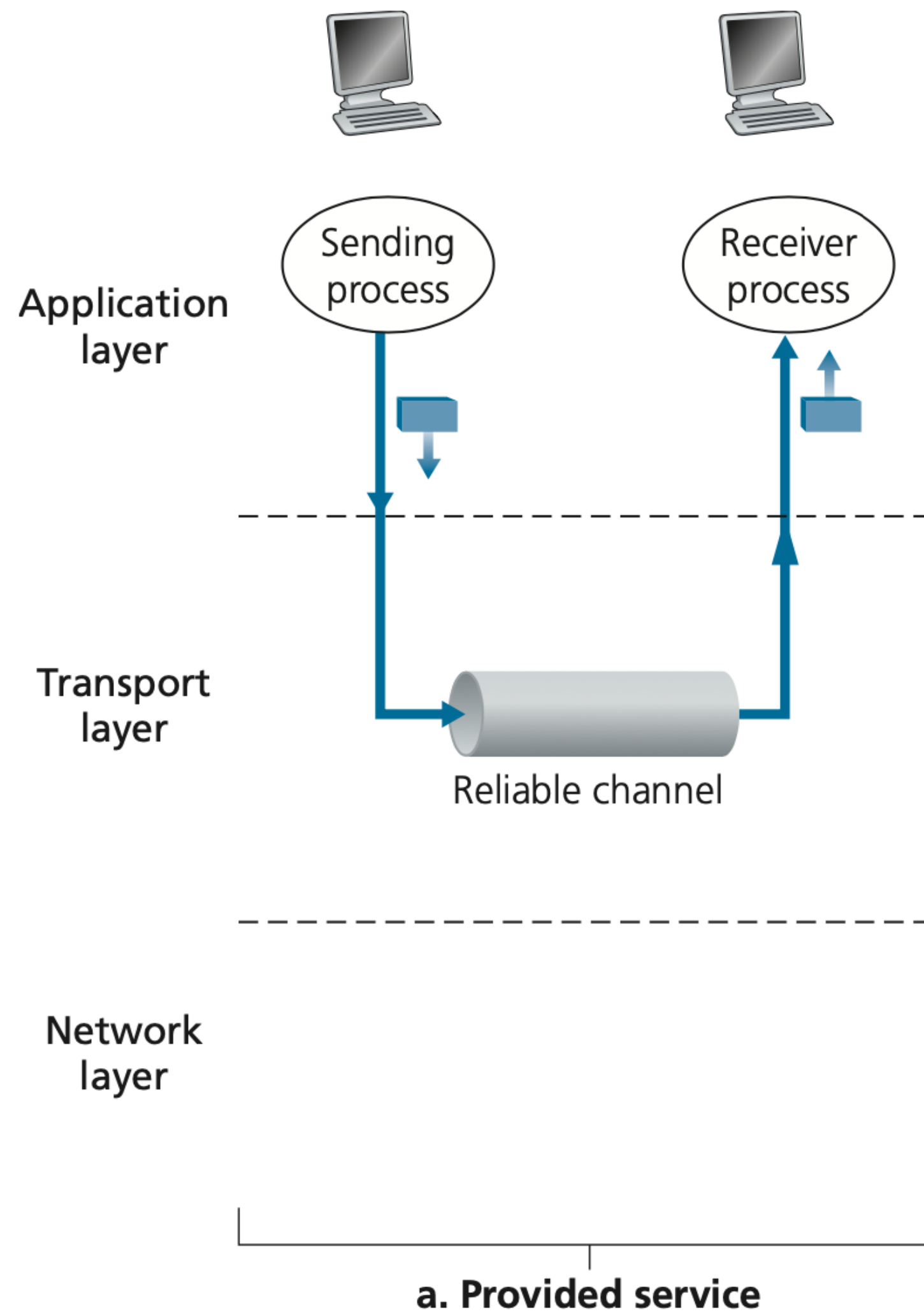
FIGURE 6.7

TCP three-way handshake.

Source: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Reliable data transfer

- **Assumptions:**
- Packets will be delivered in the order which they were sent
- The underlying channel will not reorder the packets
- Case: unidirectional data transfer



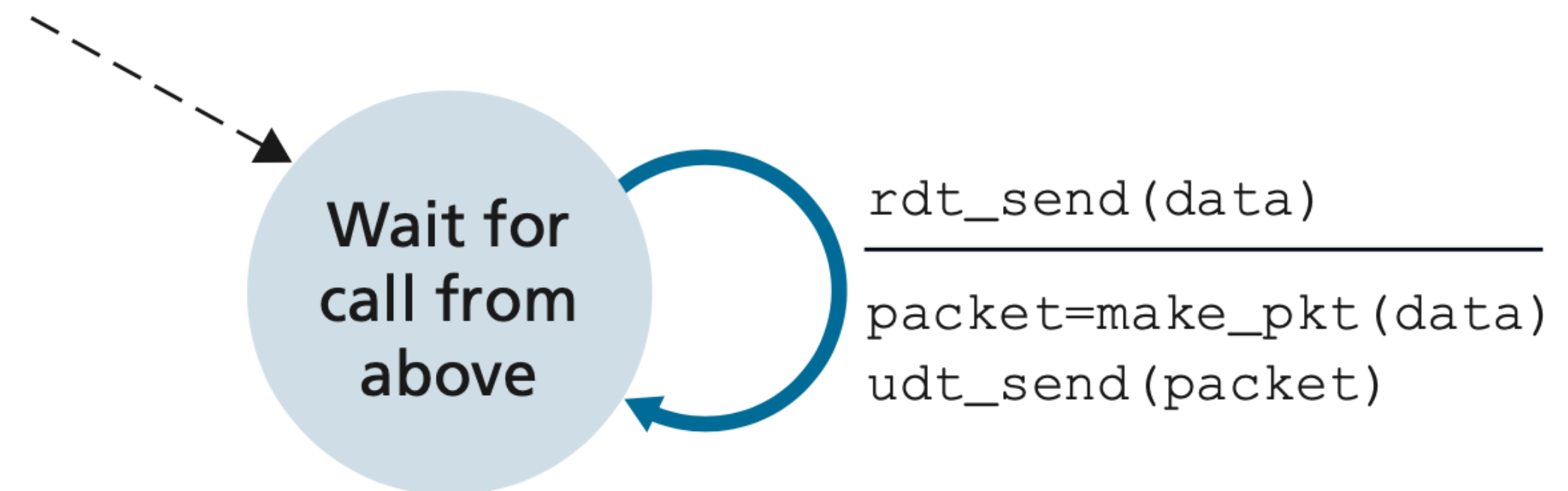
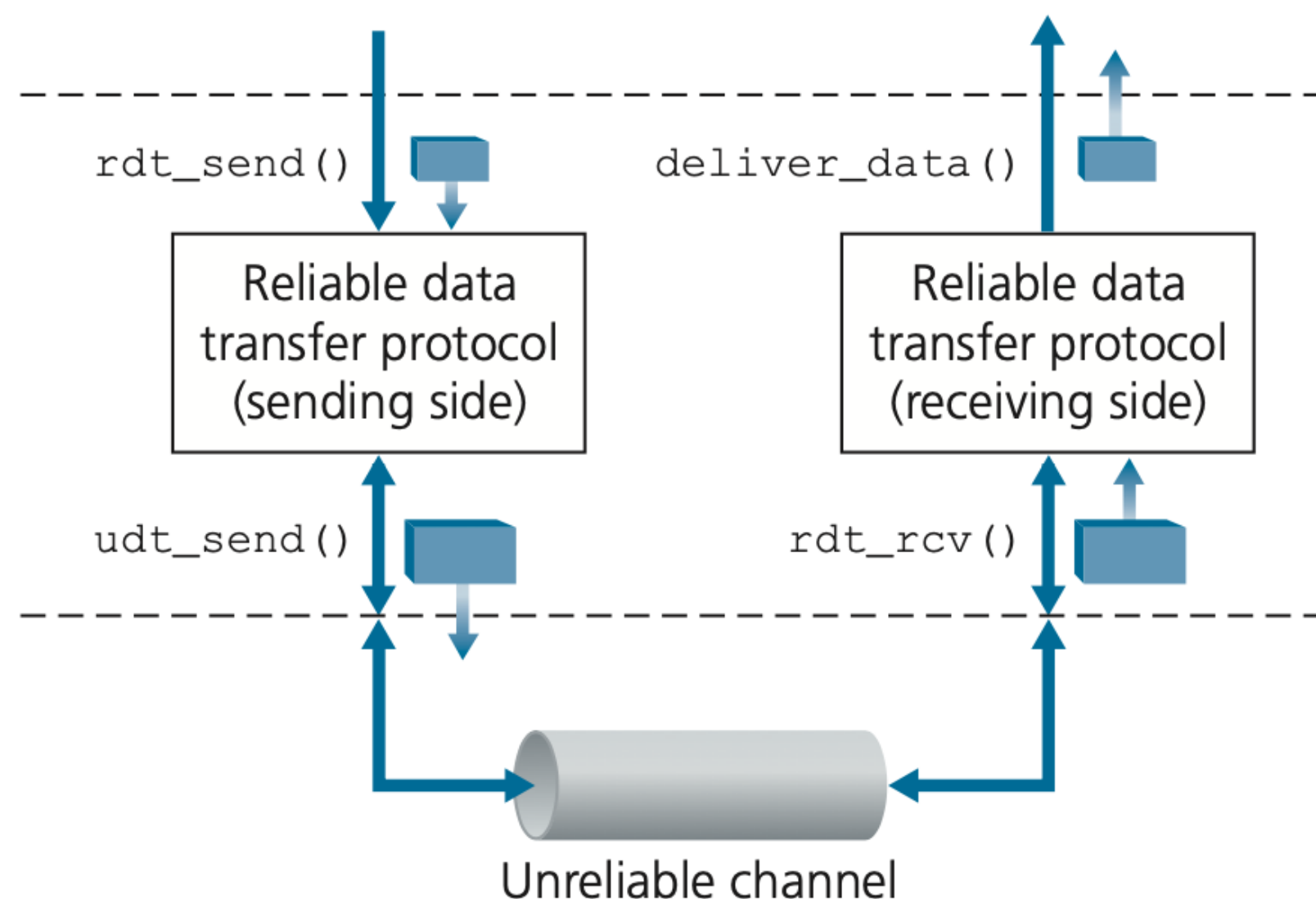
Key:

■ Data

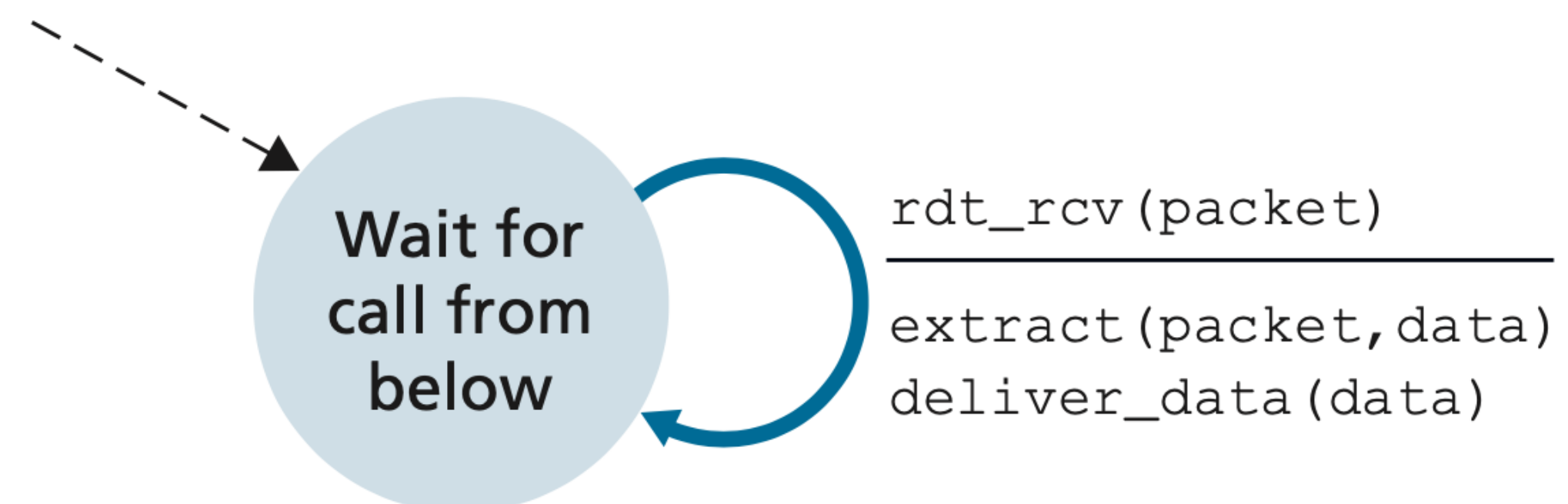
■ Packet

Finite-state machine (FSM) for rdt1.0

- rdt1.0: **Reliable Data Transfer over a Perfectly Reliable Channel**
- Sending side waits for call from above
- Receiving side waits for call from below



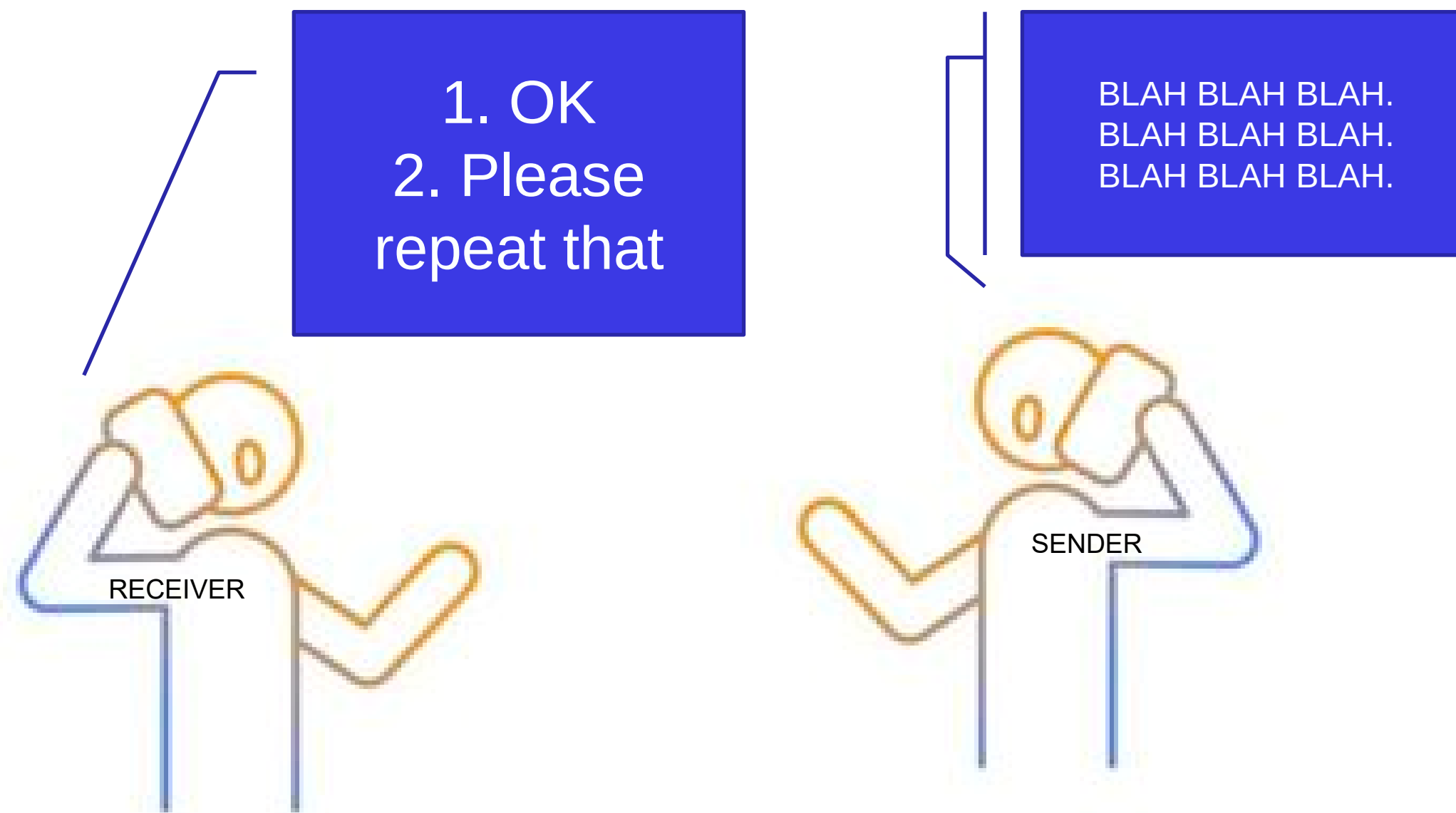
a. rdt1.0: sending side



b. rdt1.0: receiving side

Reliable Data Transfer over a Channel with Bit Errors: rdt2.0

- Acknowledgment messages
- Not acknowledged messages



ARQ: AUTOMATIC REPEAT REQUEST

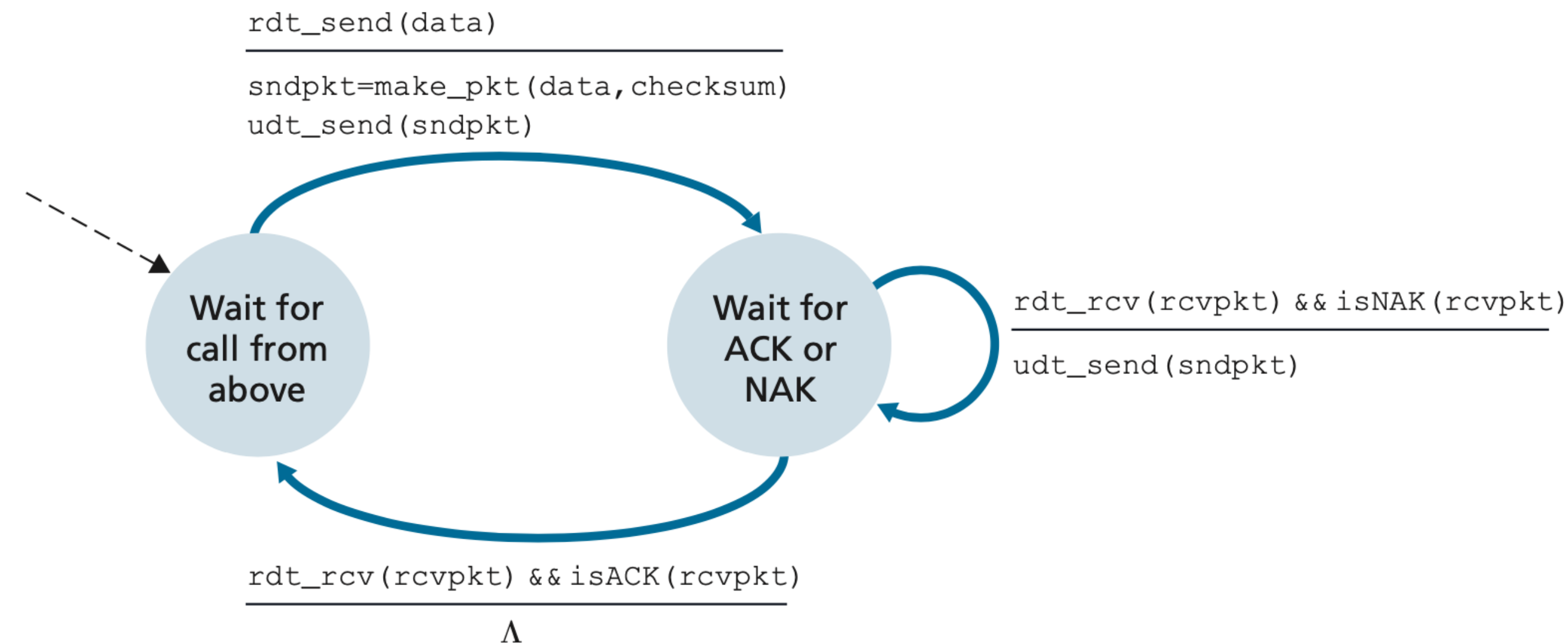
Error detection

Receiver feedback

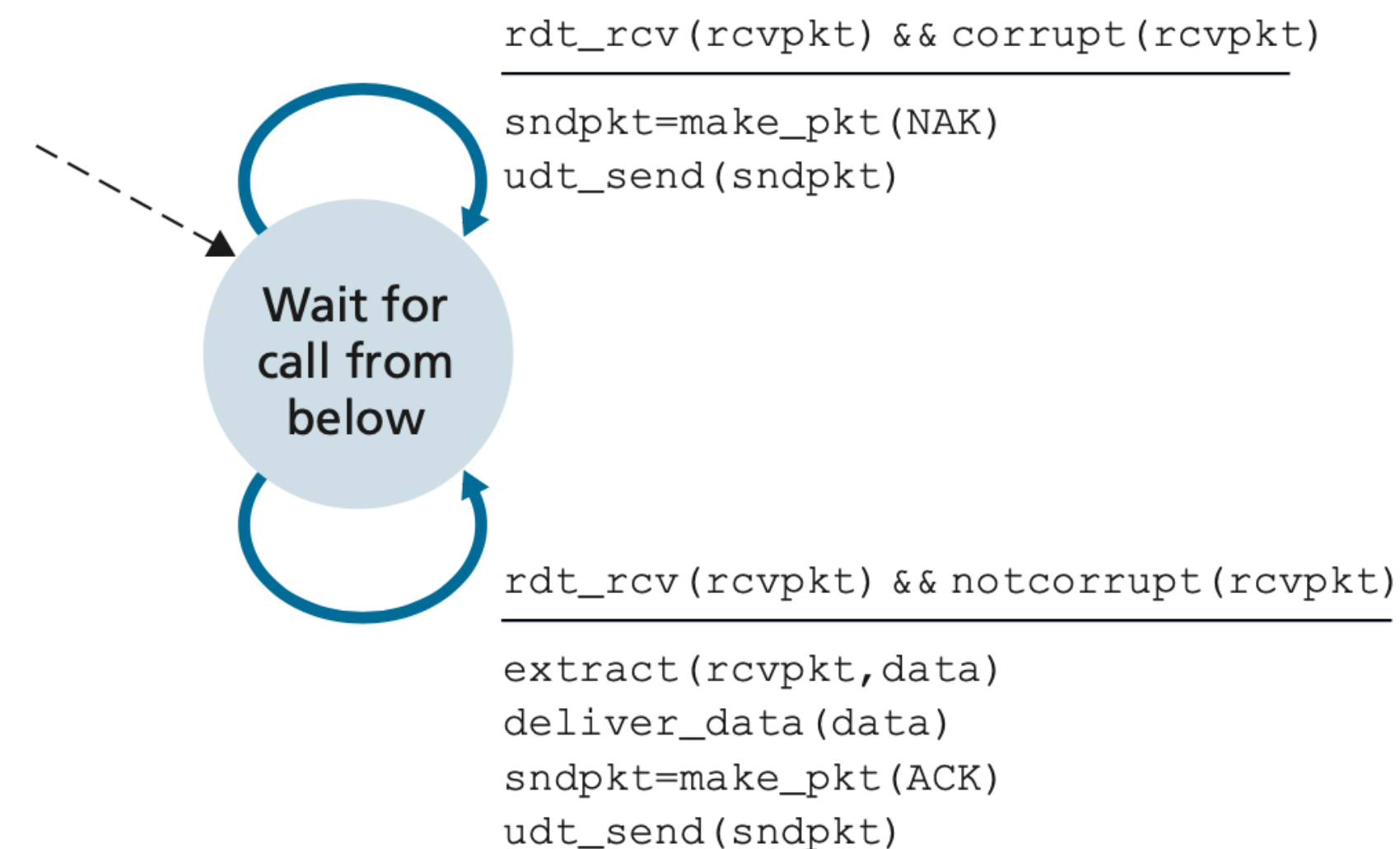
Retransmission

Reliable Data Transfer over a Channel with Bit Errors: rdt2.0

- It is important to note that when the sender is in the **wait-for-ACK-or-NAK** state, it *cannot* get more data from the upper layer: "Stop-and-wait protocol"



a. rdt2.0: sending side



b. rdt2.0: receiving side

Outline

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UDP for IoT

- Low overhead for header size:
 - packet transmissions and receptions consume less energy
 - each packet has more room for application layer data.

UDP for IoT

- Fits well with applications which demands low reliability
 - Reliability can be provided at the application layer, but this increases the complexity of the application.

TCP for IoT

- Reliable mechanism that automatically retransmits lost packets
 - for many applications, high throughput is not the primary objective and reliable delivery of data is more important.
- Complex protocol
- Large header

Outline

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Introduction

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Conclusion

- We remembered basic concepts of Networking
- We brought forward Transport-layer concepts and applications that can be used on an IoT sensor network.